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Mind-altering microorganisms: the impact of the gut microbiota on brain and behaviour

[John F. Cryan](#)  & [Timothy G. Dinan](#)[Nature Reviews Neuroscience](#) **13**, 701–712 (2012) [Save article](#)[View saved research](#) >101k Accesses | 4399 Citations | 650 Altmetric | [Metrics](#)

Key Points

- There is a growing appreciation that the gut microbiota plays a key role in maintaining homeostasis and that a disruption in its composition contributes to various disease states, including CNS disorders.
- The concept of a microbiota–gut–brain axis, although debated, is emerging to capture the importance that the microbiota has on regulating bidirectional gut–brain communication pathways.
- It is clear that stress, including stress in early life, can alter microbiota composition and this can have marked consequences on physiology in adulthood.
- Studies in germ-free animals and in animals exposed to pathogenic bacterial infections, probiotic bacteria or antibiotic drugs suggest a role for the gut microbiota in the regulation of anxiety, mood, cognition and pain.
- Although not as conceptually or empirically developed, the gut microbiota has also been implicated in obesity, autism and multiple sclerosis.
- Mechanisms as to how the microbiota are affecting gut–brain signalling are only now being unravelled. These mechanisms may include alterations in microbial composition, immune activation, vagus nerve signalling, alterations in

tryptophan metabolism, production of specific microbial neuroactive metabolites and bacterial cell wall sugars.

- Harnessing such mechanisms may pave the way for microbial-based therapeutics for various CNS disorders.

Abstract

Recent years have witnessed the rise of the gut microbiota as a major topic of research interest in biology. Studies are revealing how variations and changes in the composition of the gut microbiota influence normal physiology and contribute to diseases ranging from inflammation to obesity. Accumulating data now indicate that the gut microbiota also communicates with the CNS – possibly through neural, endocrine and immune pathways – and thereby influences brain function and behaviour. Studies in germ-free animals and in animals exposed to pathogenic bacterial infections, probiotic bacteria or antibiotic drugs suggest a role for the gut microbiota in the regulation of anxiety, mood, cognition and pain. Thus, the emerging concept of a microbiota–gut–brain axis suggests that modulation of the gut microbiota may be a tractable strategy for developing novel therapeutics for complex CNS disorders.

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Ethics declarations

Competing interests

The authors declare no competing financial interests.

Related links

FURTHER INFORMATION

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[Human Microbiome Project](#)

Glossary

Microbiota

The collection of microorganisms in a particular habitat, such as the microbiota of the skin or gut.

Stress response

The name given to the hormonal and metabolic changes that follow exposure to a threat. It involves the activation of the hypothalamus–pituitary–adrenal axis.

Microbiome

The collective genomes of all of the microorganisms in a microbiota.

Hypothalamus–pituitary–adrenal (HPA) axis

The HPA axis is the endocrine core of the stress system. Its activation results in the release of corticotropin-releasing factor from the hypothalamus, adrenocorticotrophic hormone from the pituitary and cortisol (corticosterone in rats and mice) from the adrenal glands.

Maternal separation

A model of stress in early life. Isolation of pups from their mother in early life alters maternal behaviour upon being reunited and results in permanent changes in brain and behaviour in the offspring.

Probiotic

A living microorganism that, when ingested by humans or animals, can beneficially influence health.

Inflamm-ageing

A neologism to reflect the concept that ageing is accompanied by a global reduction in the capacity to cope with various stressors and a concomitant progressive increase in pro-inflammatory status.

Mono-association

The inoculation of germ-free animals with a specific bacterium.

Bacteriocins

Proteinaceous toxins produced by bacteria to inhibit the growth of similar or closely related bacterial strain(s).

Colonic AH neurons

The major intrinsic sensory neurons in the colon. They are termed AH owing to their common electrophysiological properties whereby action potentials are followed by prolonged and substantial after-hyperpolarizing (AH) potentials.

Dysbiosis

A microbial imbalance on or within the body, often localized to the gut.

Colorectal distension

A method for assessing visceral hypersensitivity. It is a noxious visceral stimulus that can be used in studies performed in animals and humans.

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